TEXORA · COURSE NOTES · LECTURE SHEET

Human Ecology

The study of how human communities relate to their environment — the ecosystem and its energy flow, solid waste and its disposal, the pollutions of air, water and soil, and the idea of sustainable development.

HUM-2209 · SOCIOLOGY AND HUMAN ECOLOGY · TOPIC 5 · LEVEL 2 TERM 2

1. Human ecology and the ecosystem

Ecology is the study of organisms in relation to their environment; **human ecology** applies the same idea to human communities — how a population organises itself in space, draws its living from a habitat, and is changed by that habitat in return. The idea entered sociology through the Chicago School (Park and Burgess), who watched neighbourhoods of a growing city compete and settle like species in a habitat; it has since widened into the study of society–environment relations as a whole.

The working unit is the **ecosystem**: a community of living organisms (*biotic* components) together with the non-living surroundings (*abiotic* components — sunlight, water, soil, air, temperature) with which they exchange energy and materials. A pond, a paddy field, a mangrove forest and a city district can each be studied as an ecosystem. Its living members fall into three functional groups:

- **Producers** — green plants, which capture sunlight and build food by photosynthesis; every other member ultimately lives off them.
- **Consumers** — animals (and humans) that eat producers (herbivores) or other consumers (carnivores); humans are omnivores feeding at several levels.
- **Decomposers** — bacteria and fungi that break down dead matter and return its nutrients to soil and water, closing the cycle.

2. Energy in the ecosystem

Energy enters almost every ecosystem as sunlight and flows in one direction only: captured by producers, passed to consumers level by level (each feeding level is a **trophic level**), and lost as heat at every step. The chain of who eats whom is a **food chain**; real ecosystems run many interlocking chains — a **food web**. Raymond Lindeman's famous **ten per cent law** states that only about a tenth of the energy of one trophic level becomes body material at the next; the rest is spent in living or lost as heat. This is why food chains are short, why there are always fewer tigers than deer, and why eating lower on the food chain feeds more people from the same land.

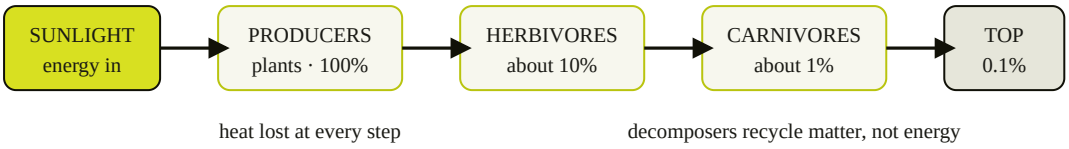


Figure 1 — One-way energy flow through trophic levels, with the roughly tenfold loss at each transfer (Lindeman's ten per cent law, approximate).

The physical environment — climate, soil, rivers — sets the terms on which all this runs. Two textbook ideas complete the picture: the **habitat**, the place an organism lives, and its **niche**, the role it plays there; and **carrying capacity**, the population an environment can support indefinitely without degrading. A human community that exceeds its habitat's carrying capacity — draining its wetlands, exhausting its groundwater — borrows from its own future, which is the seed of the problems in sections 4 and 5.

3. Solid waste and its disposal

Solid waste is the discarded solid material of households, commerce, institutions, streets and industry. Its main streams are municipal waste (food, paper, plastic, ash), industrial waste (which in a textile district includes fabric cuttings, sludge and chemical containers), hazardous waste (hospital, chemical and battery waste that can poison), and electronic waste (discarded phones, computers and boards, rich in both value and toxins).

METHOD	HOW IT WORKS	MERITS AND PROBLEMS
Open dumping	Waste tipped on open land, the default in fast-growing towns	Cheapest; but breeds disease vectors, smells, leaches poisons into soil and water, and burns spontaneously
Sanitary landfill	Waste layered, compacted and covered, on a lined site with leachate control	Safe if engineered and managed; needs land, which crowded cities lack
Incineration	Burning at high temperature, greatly reducing volume; energy can be recovered	Costly; air emissions must be controlled; ash still needs disposal
Composting	Rotting the organic fraction into manure	Ideal for the largely organic waste of South Asian cities; needs sorting at source and a market for compost

Recycling and reuse	Recovering paper, plastic, metal, glass — and textile off-cuts — as raw material	Saves resources and energy; depends on sorting, prices and honest collection chains

The governing principle everywhere is the waste hierarchy: **reduce** first, then **reuse**, then **recycle** — disposal is the last resort, not the system itself.

4. Pollution: air, water and soil

Pollution is the introduction of harmful substances or energy into the environment at a rate it cannot absorb, harming living things and human health. The three great media:

MEDIUM	MAIN CAUSES	MAIN EFFECTS
Air	Burning of fossil fuels in vehicles, industry and brick kilns; dust from construction and roads; open burning of waste; household smoke	Respiratory and heart disease; acid rain damaging crops, buildings and water bodies; greenhouse gases driving climate change
Water	Untreated industrial effluent, municipal sewage, agricultural fertiliser and pesticide run-off, oil spills	Spread of water-borne disease; death of fish and aquatic life (rivers near industrial clusters are the standing local example); poisoned drinking water and irrigation
Soil	Excess chemical fertiliser and pesticide, dumping of industrial and electronic waste, leakage of oil and chemicals	Loss of soil fertility and soil life; toxins entering food chains; land made unfit for farming or housing

For a textile engineer the water row deserves a second look: wet processing is thirsty and chemical-laden work, and the difference between a factory with a working effluent treatment plant and one without is written in the colour and smell of the nearest river. Pollution control is therefore not an ornament on this profession but part of its core duty — a point the final section generalises.

5. Sustainable development

The Brundtland Commission (1987) gave the definition every textbook repeats: **sustainable development** is *"development that meets the needs of the present without compromising the ability of future generations to meet their own needs."* It stands on three pillars that must be held together — **economic** growth that provides livelihoods, **social** fairness in who shares the benefits and burdens, and **environmental** protection of the systems life depends on. Sacrificing any pillar to the other two

eventually collapses the whole structure: growth that poisons its own water is not development but liquidation of assets.

In practice the idea asks for: using renewable resources no faster than they regrow; keeping pollution within the environment's absorptive capacity; efficiency — more value from less material and energy; and fairness between generations and within the present one. The United Nations' Sustainable Development Goals (2015) translate this into seventeen shared targets, from clean water to responsible consumption and production. For industry, and for the textile industry specifically, the working tools are familiar: effluent treatment, water and energy saving, safer chemicals, recycling of waste streams, and honest measurement of all of them.

6. Glossary — key terms of this topic

TERM	MEANING
Ecosystem	A community of organisms together with their non-living environment, exchanging energy and materials.
Trophic level	A feeding level in an ecosystem: producers, then consumers of successive orders.
Ten per cent law	Lindeman's rule that only about a tenth of one trophic level's energy passes to the next.
Carrying capacity	The population an environment can support indefinitely without degradation.
Leachate	The poisonous liquid that drains from decomposing waste into soil and groundwater.
Effluent	Liquid waste discharged by industry or sewage systems into the environment.
Sustainable development	Development meeting present needs without compromising future generations' ability to meet theirs (Brundtland, 1987).

7. Key points and summary

- An ecosystem joins producers, consumers and decomposers to their physical environment; energy flows one way through trophic levels, shrinking roughly tenfold at each step, while matter is recycled.
- Habitat, niche and carrying capacity describe how a community fits — or overflows — its environment.
- Solid waste demands the hierarchy reduce → reuse → recycle → safe disposal; open dumping is the failure the hierarchy exists to replace.

- Air, water and soil pollution share one logic: harmful inputs beyond the environment's capacity to absorb, with effects on health, livelihoods and climate.
- Sustainable development binds economy, society and environment into one obligation to the future — the standard against which industrial practice, including textile wet processing, is now judged.

8. Practice

Answer in your own words; brief model answers follow.

1. Draw the energy path grass → grasshopper → frog → snake and apply the ten per cent law qualitatively: why can the land support many frogs but few snakes?
2. “Decomposers recycle matter but not energy.” Explain why this makes sunlight, not soil, the ecosystem's true fuel.
3. A town's waste is roughly two-thirds food and garden waste. Choose the two most suitable disposal methods from this sheet and justify the choice.
4. An untreated dye-house drain enters a river. Trace three separate effects, one each on human health, on the river ecosystem, and on downstream livelihoods.
5. Define sustainable development (Brundtland). Then judge: a factory doubles output by doubling untreated discharge. Which pillars has it served and which betrayed?
6. Explain carrying capacity using a city's water supply as the example.

Answer key

1. Each transfer keeps only about a tenth of the energy: grass holds the most, grasshoppers about a tenth of that, frogs a tenth of the grasshoppers' share, snakes a tenth of the frogs'. Snakes therefore need the production of a huge base of grass and many frogs beneath them; numbers must shrink at the top.
2. Decomposers return nutrients (matter) to soil and water so producers can build again — but the energy used at every step has escaped as heat and cannot be recovered. The cycle of matter can only keep turning because new energy pours in continually from the sun.
3. Composting — the organic majority becomes manure, cheaply and locally. Recycling — for the sorted remainder (plastic, metal, paper). Landfill only for the residue. Open dumping rejected: leachate and disease from rotting organics; incineration wasteful for wet organic waste.
4. Health: contaminated drinking and bathing water spreads skin and water-borne disease. Ecosystem: chemicals and oxygen demand kill fish and aquatic organisms, collapsing the food web. Livelihoods: fishing catches fall and irrigation water damages crops, so downstream families lose income and food.

5. Development that meets present needs without compromising future generations' ability to meet theirs. The factory served the economic pillar short-term only; it betrayed the environmental pillar (poisoned water) and the social pillar (costs dumped on neighbours and the future) — so by the definition it is not sustainable development at all.
6. A city can support indefinitely only the population its renewable water supply can serve; beyond that, groundwater falls and quality degrades — the population has exceeded carrying capacity even though people continue to arrive, because the resource base itself is being consumed.

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